

ALLEN TU

Ph.D. Student, Computer Science — University of Maryland

Research Areas: Reliable and Scalable Vision Systems, 3D/4D Reconstruction, Biometrics, Generative Modeling

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SELECTED PUBLICATIONS

1. **Allen Tu***, H. Ying*, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, ‘[SpeeDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping](#)’. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
2. P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, ‘[SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting](#)’. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
3. **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, ‘[TransFIRA: Transfer Learning for Face Image Recognizability Assessment](#)’. *IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, 2026.
4. A. Hanson*, **Allen Tu***, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, ‘[PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting](#)’, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 5949–5958.
5. A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, ‘[Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives](#)’, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 21537–21546.

* denotes equal contribution.

EDUCATION

University of Maryland, College Park

Ph.D. in Computer Science

College Park, MD

January 2025 – May 2028 (Expected)

- Advised by Professor Tom Goldstein; collaborating with Matthias Zwicker, Vishal M. Patel, and Rama Chellappa

B.S./M.S. in Computer Science, Minor in Statistics

August 2019 – December 2024

RESEARCH EXPERIENCE

University of Maryland Institute of Advanced Computer Studies

Graduate Research Assistant

College Park, MD

August 2023 – Present

- Introduced pruning, rasterization, and motion distillation methods that accelerate static and dynamic 3D Gaussian Splatting, producing representations with over $10\times$ fewer primitives while preserving visual fidelity [1, 4, 5]
- Incorporated uncertainty-aware diffusion and super-resolution priors into 3D reconstruction pipelines to improve fidelity under sparse and low-resolution supervision while mitigating artifacts and cross-view inconsistencies [2]
- Designed scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude ([WRIVA](#)) program

Systems & Technology Research

Computer Vision Research Intern

Arlington, VA

June 2022 – January 2026

- Introduced an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [3]
- Demonstrated state-of-the-art template-based recognition on the proprietary BRIAR surveillance benchmark, improving TAR from 0.43 to 0.87 at $1e-3$ FMR using an encoder trained solely on public data [3]
- Developed multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range ([BRIAR](#)) program

SERVICE

Workshop Organizer: [SPAR-3D: Security, Privacy, & Adversarial Robustness in 3D Generative Vision Models](#), CVPR 2026